NATIONAL HIGHER SCHOOL OF MATHEMATICS
المدرسة الوطنية العليا في الرياضيات

1^{st} **year**
Discrete Mathematics I
2023/2024

Mid-term exam solution

Exercice 1 (7.5 pts). *There are 6 boys and 5 men waiting for their turn in a barber shop. Two particular boys are A and B, and one particular man is Z. There is a row of 11 seats for the customers. Find the number of ways of arranging them in each of the following cases:*

1. *there are no restrictions;*
2. *A and B are adjacent;*
3. *Z is at the centre, A at his left and B at his right (need not be adjacent);*
4. *boys and men alternate.*
5. *Z is at the centre and adjacent to A and B;*
6. *all men form a single block;*
7. *no two of A, B and Z are adjacent;*
8. *all boys form a single block and Z is adjacent to A;*

Solution:

1. This is the number of permutations of the 11 persons. The answer is 11 !.
2. Treat $\{A, B\}$ as a single block. The number of ways to arrange the remaining 9 persons together with this block is $(9+1)$!. But A and B can permute themselves in 2 ways. Thus the total desired number of ways is, by (MP), $2 \cdot 10!$.
3.

					Z					

As shown in the diagram above, A has 5 choices and B has 5 choices as well. The remaining 8 persons can be placed in 8 ! ways. By (MP), the total desired number of ways is $5 \cdot 5 \cdot 8$!.

4. The boys (indicated by b) and the men (indicated by m) must be arranged as shown below.

b	m	b	m	b	m	b	m	b	m	b

The boys can be placed in 6 ! ways and the men can be placed in 5 ! ways. By (MP), the desired number of ways is $6! \cdot 5!$.

5. Put Z in the center. There is only one way of doing this.

   Put A and B adjacent to Z. There are 2 ways of doing this.

   Arrange the remaining 8 persons. There are 8 ! ways of doing this.

   Thus, the number of ways $= 2 \cdot 8!$.

6. Consider the 5 men together as a single block. Now, there are 7 blocks, i.e. the 6 boys and the block, to arrange in a row. There are 7! ways of arranging 7 blocks.

   and there are 5! to arrange the men between themselves.

   Thus, the number of ways is $7! \cdot 5!$.

7. 1^{st} **appraoch:**

   First, arrange the other 8 persons in a row. There are 8 ! ways of doing this.
   To ensure that no two of A, B and Z are adjacent, we now put each of them one by one into the spaces between the other 8 persons (see figure below).
   Number of choices for $A = 9$.
   Number of choices for $B = 8$.
   Number of choices for $Z = 7$.
   Thus, the number of ways $= 8! \cdot 9 \cdot 8 \cdot 7$.

$$\begin{array}{ccccccccccccccccc} \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ & \text{X} & & \text{X} & & \text{X} & & \text{X} & & \text{X} & & \text{X} & & \text{X} & & \text{X} & \end{array}$$

   2^{nd} **appraoch:**
   Let us consider the three particular persons as separators and we arrange the remaining people, then we express the number of ways to do that as an equation and the number of solution of this equation is the same number of ways to arrange the people such that Z, A and B are not adjacent.

$$\begin{cases} x_1 + x_2 + x_3 + x_4 = 8, \\ \text{where } x_1 \geq 0, x_2 \geq 1, x_3 \geq 1 \text{ and } x_4 \geq 0. \end{cases}$$

   Making a change of variable, we get

$$\begin{cases} x'_1 + x'_2 + x'_3 + x'_4 = 6, \\ \text{where } x_i \geq 0, \text{for } i \in \{1, 2, 3, 4\}. \end{cases}$$

   The number of solution of the equation is given by

$$\left(\!\!\binom{4}{6}\!\!\right) = \binom{9}{7} = \binom{9}{3},$$

   we should add the order of the three particular persons which is 3! and the remaining people which is 8! , by (MP) we have

$$8! \binom{9}{3} 3! = 8! \ (9)_3 = 8! \cdot 9 \cdot 8 \cdot 7$$

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8. First, consider the 6 boys together as a single block, with Z adjacent to A, as a single block. There are 2 ways to do this, i.e. $ZAXXXXX$ or $XXXXXAZ$.

   Now, there are 5 blocks, i.e. the block and the other 4 men, to arrange in a row. There are 5! ways of arranging 5 blocks.

   the 5 boys and arrange them. (Note that A and Z are fixed.)

   There are 5 ! ways of doing this.

   Thus, the number of ways $= 2 \cdot 5! \cdot 5!$.

Exercice 2 (6.5 pts). *Find the number of ways of forming a group of $2k$ people from n couples, where $k, n \in \mathrm{N}$ with $2k \leq n$, in each of the following cases:*

1. *There are k couples in such a group;*
2. *No couples are included in such a group;*
3. *At least one couple is included in such a group;*
4. *Exactly two couples are included in such a group.*

Solution:

1. The number of ways to choose a group of $2k$ people from n couples, given that there are k couples in such a group, is equal to the number of ways to choose k couples out of n, which is equal to $\binom{n}{k}$.

2. Firstly, there are $\binom{n}{2k}$ ways to choose $2k$ couples out of n. Then for each chosen couple, there are 2 ways to choose one person out of the couple. By (MP), the number of ways is equal to $\binom{n}{2k} \cdot 2^{2k}$.

3. By definition, there are $\binom{2n}{2k}$ ways to choose $2k$ people out of n couples. Then there are $\binom{2n}{2k} - \binom{n}{2k} \cdot 2^{2k}$ ways to ensure that at least one couple is included in such a group.

4. Firstly, there are $\binom{n}{2}$ ways to choose 2 couples out of n. Then there are $\binom{n-2}{2k-4}$ ways to choose $2k-4$ couples out of the remaining $n-2$ couples. Then for each of the remaining $2k-4$ couples, there are 2 ways to choose one person out of the couple. By (MP), the number of ways is equal to $\binom{n}{2}\binom{n-2}{2k-4} \cdot 2^{2k-4}$.

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Exercice 3 (6 pts).

- *Determine the number of integer solutions for the equation:* $\begin{cases} x_1 + x_2 + \ldots + x_k = r, \\ \textit{where } x_i \geq 0, \textit{for } i \in \{1, \ldots, k\}. \end{cases}$

- *Find the number of integer solutions for the inequality* $\begin{cases} x_1 + x_2 + \ldots + x_r < n, \\ \textit{where } x_i \geq 0, \textit{for } i \in \{1, \ldots, r\}. \end{cases}$

- *Determine the number of integer solutions for the equation:* $\begin{cases} |x_1| + |x_2| + \ldots + |x_n| = k \\ \textit{where } |x_i| \geq 1, \textit{for } i \in \{1, \ldots, n\}. \end{cases}$

- *Explain why the equation:* $\begin{cases} |x_1| + |x_2| + \ldots + |x_n| = k \\ \textit{where } |x_i| \geq 0, \textit{for } i \in \{1, \ldots, n\}. \end{cases}$

  has exactly $\sum_{r=0}^{n} \binom{n}{r} 2^{n-r} \left(\!\!\binom{n-r}{k-n+r}\!\!\right)$ *integer solutions.*

Solution:

- The number of solutions for $x_1+x_2+\ldots+x_k = r$ with $x_i \geq 0$ is given by the the combination with repetition:

$$\left(\!\!\binom{k}{r}\!\!\right) = \binom{r+k-1}{r} = \binom{r+k-1}{k-1}.$$

  This represents the number of ways to arrange $(k-1)$ separators $'|'$ and r $'1'$s in a row, where each arrangement corresponds to a valid solution to the equation. The separators divide the row into k sections, and the number of $'1'$s in each section represents the value of x_i.

- By adding an extra variable x_{r+1} we transform the inequality to an equality,

$$\begin{cases} x_1 + x_2 + \ldots + x_r + x_{r+1} = n, \\ \text{where } x_i \geq 0, \text{for } i \in \{1, \ldots, r\},\ x_{r+1} \geq 1. \end{cases}$$

  We make a change of variables to set all the variables greater or equal to 0.

$$\begin{cases} x'_1 + x'_2 + \ldots + x'_{r+1} = n-1, \\ \text{where } x'_i \geq 0, \text{for } i \in \{1, \ldots, r+1\}. \end{cases}$$

  Now, the nubmer of solution of the equality is given by

$$\left(\!\!\binom{r+1}{n-1}\!\!\right) = \binom{r+n-1}{n-1} = \binom{r+n-1}{r}$$

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- As in the first question, the count of positive solutions is $\left(\!\!\binom{n}{k-n}\!\!\right) = \binom{k-1}{n-1}$. For each value of x_i, the negative counterpart also provides a solution. Thus, each variable has two choices, resulting in 2^n possibilities.

  Consequently, the overall number of solutions is $2^n \cdot \binom{k-1}{n-1}$.

- To understand why the given equation has exactly

$$\sum_{r=0}^{n} \binom{n}{r} 2^{n-r} \left(\!\!\binom{n-r}{k-n+r}\!\!\right)$$

  integer solutions, we can break down the reasoning:

  - $\binom{n}{r}$ ways to choose r variables to be zeros.
  - $2^{n-r} \left(\!\!\binom{n-r}{k-n+r}\!\!\right)$: the number of solutions of the remaining equation which is the same as the previous equation

$$\begin{cases} |x_1| + |x_2| + \ldots + |x_{n-r}| = k \\ \text{where } |x_i| \geq 1, \text{for } i \in \{1, \ldots, n-r\}. \end{cases}$$

  - Summation over r: The final expression is the sum over all possibilities, considering different values of r for the number of zeros.

Exercice 4 (Optional. **1.5 pts**).

1. *Prove in two different ways the following identity* $k\binom{n}{k} = n\binom{n-1}{k-1}$.
2. *Calculate* $\sum_{r=0}^{n} 3^r \binom{n}{r}$, *and then give a combinatorial proof for the derived identity.*

Solution:

1. $k\binom{n}{k} = n\binom{n-1}{k-1}$;

   (a) ***Algeraic proof***:

$$k\binom{n}{k} = k\frac{n!}{k!(n-k)!} = \frac{n(n-1)!}{(k-1)!(n-k)!} = n\frac{(n-1)!}{(k-1)!(n-k)!} = n\binom{n-1}{k-1}.$$

   (b) ***Combinatorial proof***:
   The LHS, counts the number of ways to select a committee of k person from n people and we have k possibilities to choose a leader within this committee. This clearly equivalent to the

number of ways to choose a leader from a set of n people and then the remaining $k-1$ members of the committee from the remaining $n-1$ people, which is the RHS.

2. *(a)* $\sum_{r=0}^{n} 3^r \binom{n}{r} = (1+3)^n = 4^n.$

   (b) *The LHS, counts the number of ways to form a r-member committee, from n people and then affect them to 3 tasks (each one have 3 choices), so we have 3^r to affect the r chosen people. Thus, the number of ways to form a r-member committee including the affectation of tasks is, by the multiplication principle, $3^r\binom{n}{r}$. As r could be $1, 2, \ldots, n$, by the addition principle, the number of ways to do so is given by $\sum_{r=0}^{n} 3^r\binom{n}{r}$. This is equivalent to the fact that each one of n people have 4 choices (3 if the person is chosen and then affected to tasks and 1 if the person is not affected to any task) thus the number is 4^n, which is the RHS.*